

A physician measures the arterial blood pressure of a patient while she is resting and again while she is running on a treadmill. The results of these blood pressure tests are given in the table below.

Activity	Blood pressure (mmHg)
Resting	128 / 80
Running	155 / 73

Given this data, was arterial blood pressure during heart contraction higher while the patient was resting or while she was running?

- ☐ A. Resting, because her diastolic blood pressure was higher when at rest. [6%]
- ☐ B. Resting, because her systolic blood pressure was higher when at rest. [2%]
- ✓ ☒ C. Running, because her systolic blood pressure was higher while running. [86%]
- ☐ D. Running, because her diastolic blood pressure was higher while running. [4%]

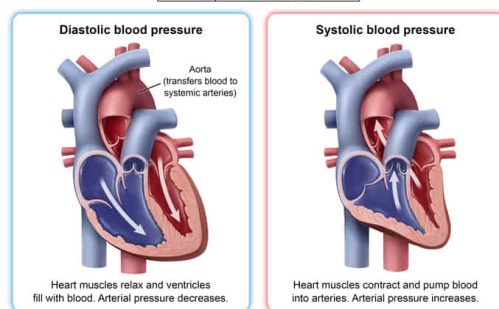
Correct

85% answered correctly

Explanation:

Systolic vs diastolic blood pressure

Activity	Blood pressure (mmHg)
Resting	128 / 80
Running	155 / 73



Contractions of the **heart** pump blood through **vessels** of the **circulatory system**, facilitating the delivery of oxygen and nutrients to tissues and the removal of waste products (eg, carbon dioxide). Specifically, blood moves away from the heart through the arteries and returns to the heart via the veins. The repetitive heart contractions that facilitate this transport of blood occur in two distinct phases:

- During the **diastole** phase, the heart *relaxes* and fills with blood returning to the heart via the veins. Because blood is *not* pumped from the heart during diastole, the volume of blood flowing through the arteries is *lower*, **decreasing pressure** on the arterial walls. Therefore, arterial blood pressure is *lowest* during diastole and is measured as the **diastolic blood pressure**.
- During the **systole** phase, the **heart contracts** and blood is forced from the heart into the arteries. This *increased* volume of blood flowing through the arteries on contraction of the heart **exerts increased pressure** (ie, hydrostatic pressure) on the arterial walls. Accordingly, arterial blood pressure is *highest* during systole and is measured as the **systolic blood pressure**.

In the given scenario, a physician measures the arterial blood pressure of a patient while she is resting and while she is running on a treadmill. Generally, blood pressure measurements (mm Hg)

are represented by the higher systolic pressure *over* the lower diastolic pressure. Therefore, the patient's blood pressure on heart contraction (systolic blood pressure), as shown in the table, was **higher while running (155 mm Hg)**.

(Choices A and D) Diastolic blood pressure is measured within the arteries on *relaxation* of the heart. Although the patient's diastolic blood pressure was *higher* while resting, this value is *not* representative of her blood pressure during heart contraction (ie, systolic blood pressure).

(Choice B) Systolic blood pressure (measured during heart contraction) is generally *greater* than diastolic blood pressure and is represented by the upper number in each measurement. Therefore, systolic blood pressure was *lower* while the patient was resting.

Educational objective:

Arterial blood pressure is highest when the heart contracts (during systole) and lowest when the heart relaxes (diastole). When measuring blood pressure (mm Hg), the higher systolic blood pressure is recorded over the lower diastolic blood pressure.

Time spent:0 sec

Subject:
Biology

QID:402416

System:
Circulation and Respiration